Political Geography 110 (2024) 103077

Contents lists available at ScienceDirect

Political Geography

journal homepage: www.elsevier.com/locate/polgeo

Full Length Article

Where do donors come from? Using census data to predict donations to Canadian federal election candidates

Holly Ann Garnett [a,b,*]

[a] Department of Political Science & Economics, Royal Military College of Canada, PO Box 17000, Station Forces, Kingston, Ontario, K7K 7B4, Canada
[b] University of East Anglia, UK

ABSTRACT

Who is mostly likely to contribute to electoral-district level candidates in Canadian federal elections? This paper uses data from the Elections Canada Political Financing Database of Contributions made to individual candidates in the 2015, 2019 and 2021 general elections, matched to socio-demographic data from the 2016 census, at the Dissemination Area level. We find that the areas that are most likely to have donors are older, with higher median income and higher percentage of the population with post-secondary education. The data also further supports previous findings from the Canadian context suggesting that areas with a greater percentage of visible minorities of South and Southeast Asian origin are most likely to donate to local campaigns. When turning to overall contribution amounts, we find that the effects of these socio-demographic variables are somewhat weakened (though not eliminated). This points to a major difference between areas producing donors vs. those without donors, rather than between the levels of donations in a small geographic area.

To run in a federal election in Canada, there are several formal and informal requirements a potential candidate must fulfill: winning the party nomination (or running as an independent), setting up a campaign team, submitting their nomination papers to the district returning office with the required signatures, and raising the funds necessary to advertise, hire staff and pay for their expenses. While some percentage of expenses may be reimbursed through public subsidies (Elections Canada, 2021, pp. 121-123), most funds used for campaigns must be available up-front and reimbursed later. The candidate's party may have funds to transfer to a candidate, but they are still expected to raise funds themselves. Since corporate and union donations have been prohibited in Canada, and individual candidates can only contribute a certain amount to their own campaign (Elections Canada, 2021, p. 24), the main source of funds to finance elections is individual donations (Elections Canada, 2021, p. 24).

Research from the Canadian and American context have given us some sense of who is most likely to donate, based on socio-demographic variables (Besco & Tolley, 2022; Tolley et al., 2022), attitudes, or motivations (Garnett et al., 2022). However, much of this research relies either on self-reported survey data, which is difficult to obtain given survey non-responses rates (see Garnett et al. (2022)), or extrapolation from the names of candidates to ascertain gender or ethnicity, which can be inaccurate, especially for some ethnic groups (Mateos, 2007).

This article takes an alternative approach, using data from the Elections Canada Political Financing Database of Contributions. It focuses exclusively on officially reported donations to the district-level candidates that make up the Canadian electoral system in the 2015, 2019 and 2021 federal elections. These data are matched to socio-demographic data from the 2016 census, at the smallest geographic level reported (census dissemination area (DA)). This avoids relying on potentially less reliable sources of data such as survey responses or extrapolation of characteristics based on a donor's name.

This article ascertains the profiles of the geographic areas where donors are most likely to come from, including previously understudied characteristics – including income, education, age, and ethnicity – that define Canadian donors to local candidates. While Canadian campaign finance regulations are aimed at curbing the inordinate influence of wealthy business, 'special interests,' or rich candidates, the importance of money in politics still has a role in making the electoral system exclusionary. This article seeks to determine who is making campaign contributions to local candidates and who may be left out of this uniquely local form of political engagement in the Canadian federal electoral system.

1. Theoretical motivations: Money and democratic equality

A healthy democracy relies on popular control of the government, as realised through equal participation in the political system (Beetham, 1994; Boese & Wilson, 2022; Coppedge et al., 2008; Dahl, 1971; Wong, 2021). To achieve this end, all citizens require equal opportunities to

* Department of Political Science & Economics, Royal Military College of Canada, PO Box 17000, Station Forces, Kingston, Ontario, K7K 7B4, Canada.
E-mail address: holly-ann.garnett@rmc-cmr.ca.

https://doi.org/10.1016/j.polgeo.2024.103077
Received 13 September 2023; Received in revised form 6 February 2024; Accepted 7 February 2024
Available online 28 February 2024

contest elections, to deliberate on the future of the polity, and ultimately to participate in the mechanisms that translate votes into representation in national legislative bodies. Equal participation extends beyond the act of voting to engagement in alternative forms of participation, like signing a petition, attending a protest, or sharing information online. In each method of participation, there must remain the principle of equality, meaning that some voices or opinions are not deliberately amplified (or suppressed) over others. Ideally, any barriers to participation among those with low education or income must be reduced as much as possible.

Donors participate in the political process by 'putting their money where their mouth is,' so-to-speak. However, research demonstrates time and time again that magnifying some voices over others, based on their donating or fundraising capacity, can be a slippery slope towards creating or perpetuating inequalities (Verba, Schlozman, and Brady, 1995). Donors and non-donors rarely share the same characteristics (Garnett et al., 2022). This may be based on their socio-demographic profile—including wealth and education levels (Ansolabehere, de Figueiredo, and Snyder, 2003; Francia et al., 2003; Gimpel et al., 2006; Jacobson, 1980; James, 2009; Jones, 1990; Lipsitz & Panagopoulos, 2011; Schlozman, Verba, and Brady, 2012; Verba, Schlozman, and Brady, 1995), and gender (and the related access to financial networks and the 'purse strings') (Francia et al., 2003; Lipsitz & Panagopoulos, 2011; McMahon et al., 2021; Tolley et al., 2022)— or by feelings of political efficacy, interest, and knowledge about the political system (Ansolabehere, de Figueiredo, and Snyder, 2003).

Unfortunately, these characteristics are the same ones that predict differences in other forms of political participation, like voting. For these reasons, Verba, Schlozman, and Brady (1995) argue that political giving is perhaps the most exclusionary form of political participation, since it is overwhelmingly a function of a donor's income. Unlike forms of participation like voting and volunteering, which, while skewed towards higher income levels due to time and cognitive costs, are an option available to all voters, donating to a campaign is necessarily excluded to those with the discretionary income available. In their seminal work *Voice and Equality*, Verba, Schlozman, and Brady (1995) define resources as a key component of the civil voluntarism model. Political donations represent perhaps the most explicit way that resources influence the ability of citizens to participate in politics. Citizens who are not donors may therefore feel less engaged with the political process and, for example, less entitled to contact their local representative than those who donate to campaigns (Bartels, 2008, 2009; Schlozman, Verba, and Brady, 2012).

In many countries, wealthy citizens can magnify their voices through political donations, as evidenced by work that shows that economic elites have more influence on politics, with their preferences more commonly acted upon (Gilens & Page, 2014; Winters & Page, 2009). Money may buy access to politicians (Ansolabehere, de Figueiredo, and Snyder, 2003); conversely, an inability to donate can exclude many people from this type of network and access. While a system of patronage appointments is (we hope) of a bygone era, donating may yet have informal benefits, in terms of community prestige and access to networks. Finally, if more money can increase the likelihood of the preferred candidate winning, as has been demonstrated in Canada (Eagles, 2004), then those who are able to donate are able to also influence the makeup of parliament, with politicians who support their goals.

2. Hypotheses: The socio-demographic predictors of donating

This article therefore asks the question: What characteristics of an individual predicts their propensity to donate to electoral district candidates in Canada? Due to the limited research available from the Canadian context specifically, we must draw on Canadian, American, and international literature on the predictors of political donations. In literature in each of these contexts, some key patterns of individual predictors of donating emerge that appear common to most developed democracies.

The first, most obvious, potential predictor of political donations is an individual's ***income***. One can expect donors to be the "socio-economic elite" (Costantini & King, 1982), who have a larger amount of discretionary income that can be earmarked towards political contributions. This has been demonstrated empirically in a plethora of American studies of campaign finance (Ansolabehere, de Figueiredo, and Snyder, 2003; Francia et al., 2003; Gimpel et al., 2006; Jacobson, 1980; James, 2009; Jones, 1990; Lipsitz & Panagopoulos, 2011; Schlozman, Verba, and Brady, 2012; Verba, Schlozman, and Brady, 1995), but also studies looking at other types of donations, for example to charities (James, 2009). The small number of Canadian studies on this issue present similar findings (Carmichael & Howe, 2014). The reasons behind a relationship between income and donations could include that candidates, if coming from higher income levels themselves, would primarily ask their colleagues and friends at similar income levels for donations (Gimpel et al., 2006). Due to the focus on donations to individual candidates in this article, where it may be more likely to be asked directly to donate to a personal campaign, it is expected that this will apply in the Canadian context as well.

H1. Areas with higher median income will be more likely to have donors and will have higher levels of donations.

Along with income, ***education*** levels are commonly studied as a correlate of political participation (Lipsitz & Panagopoulos, 2011). One could expect that those with higher levels of education, particularly post-secondary education, would be more likely to make political donations because they have been cognitively mobilized to become interested in political issues through their higher education (Verba, Schlozman, and Brady, 1995). They may also be more likely to have been socialized to believe that making political donations is a normal way of participating in politics, through their family or social circles (Johnson, 2013). Like with income, those with higher education may also be more likely to be asked for political donations from their friends and colleagues who participate in politics (Gimpel et al., 2006). It is expected that this finding will hold in Canada as well, especially in the case of donations to individual candidates, where activity in political networks may be most important.

H2. Areas with higher rates of post-secondary education will be more likely to have donors and will have higher levels of donations.

There are a number of reasons why ***age*** influences the likelihood of making political donations (Lipsitz & Panagopoulos, 2011; Schlozman, Verba, and Brady, 2012). First, it is likely that one's income increases with age, as more experience is gained on the job and promotions (and raises) are earned. With higher income comes a greater pool of resources which could be donated to political campaigns. Second, there is the well-known voter apathy gap between the youngest and oldest voters (Norris, 2002; Smets & van Ham, 2013). Young people are less likely to vote due to a lack of interest and mobilization and therefore logically might be less likely to donate to campaigns. Additionally, one may expect the youngest voters to be involved in other forms of activism. For example, youth are more likely to attend a protest (Schlozman, Verba, and Brady, 2012). Drawing on scholars who argue that there is a decline in citizen activity in political parties (Everson, 1982; Wattenberg, 1984; Meisel & Mendelsohn, 1996; Cross & Young, 2006), it is also conceivable that young voters may be directing their giving towards causes other than traditional party politics. Finally, just as those with higher income and education levels tend to be targeted for political donations through their social networks, we may see a lower likelihood of young citizens donating since members of their social circles are less likely to be running for office. For this reason, it is expected that this relationship between age and donating will apply in the context of candidate donations in Canada.

H3. Areas with a higher average age will be more likely to have donors

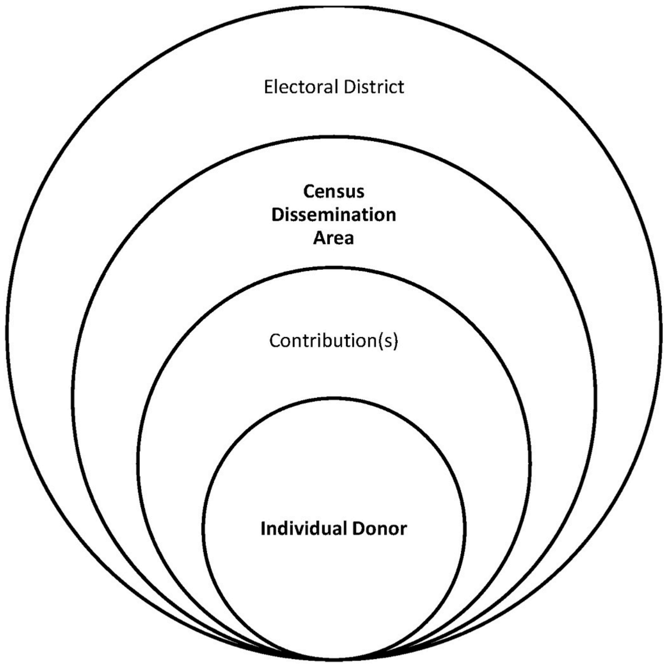

Fig. 1. Donors nested in electoral districts.

and will have higher levels of donations.

The influence of ***race and ethnicity*** on donating to political campaigns has also been discussed, especially in the American context. Scholars have found that white individuals in the United States are more likely to donate (Francia et al., 2003; James, 2009). While this may be related to the socio-economic status of minority groups, research demonstrates that African Americans and Latinx individuals tend to donate less to campaigns, even once income has been accounted for (King, 2009). By contrast, Asian Americans are no less likely to donate (Cho, 2002). Intersectional research has also shown that women of colour are far less likely to be campaign contributors in the American context (Grumbach, Sahn, and Staszak, 2022). In Canada, research by Besco and Tolley (Besco & Tolley, 2022) matching the names of donors with their suggested ethnicity finds that ethnic minority groups donate less than those with surnames of European ancestry. However, one notable exception is those of South Asian descent, who give twice as many donations than their share of the Canadian population, many directed to those of the same ethnic group.

H4a. Areas with a higher percentage of visible minorities will be less likely to have donors and will have lower levels of donations.

H4b. Areas with a higher percentage of visible minorities of South Asian decent will be more likely to have donors and will have higher levels of donations.

Finally, while there remains a robust debate surrounding ***gender*** gaps in the voter turnout among women (Cordova & Rangel, 2017; Harell, 2009; Verba et al., 1997), it is possible that the wage gap and gendered differences in control of household finances may lead women to have lower rates of political donations. This has been demonstrated in some research that has found that men are more likely to give to campaigns than women (Francia et al., 2003; Lipsitz & Panagopoulos, 2011). In the Canadian context, research by Tolley et al. (2022), considering 25 years of data, shows that women make up only one-third of political donors in Canada. Additionally, they note that women are more likely to give to national campaigns than local candidates. Looking specifically at donations during the period of the COVID-19 pandemic, McMahon et al. (2021) find similar results: that women are less likely to donate. A hypothesis regarding gender, however, cannot be tested in this article due to the use of census data at the dissemination area (discussed later in this article), which does not have much variation.

3. Methods

The cases studied are 2015, 2019 and 2021 general elections in Canada. The 2015 election was a regularly scheduled fixed-date election, following a majority Conservative government. It was the longest campaign period for a federal election in Canada at 11 weeks and was notable as a competitive race between three major parties – with each party in the lead at some point during the campaign (Ferreira, 2015). It attracted considerable interest with the highest turnout in recent years (Elections Canada, 2023c). The subsequent 2019 re-election of the majority Liberal government attracted considerably less turnout (Elections Canada, 2023c). The 2021 federal election was unique in that it took place during the COVID-19 pandemic. With support for the governing Liberals waning, they were re-elected with fewer seats and a minority parliament.

To study political donations to candidates in these three general

Table 1
Dissemination area contributions.

	2015	2019	2021
Percentage of DAs with any contribution to a district-level candidate	22%	15%	12%
Average total contributions per 100 people (Including DAs with no contributions)	$ 36.66	$ 22.99	$ 19.22
Average total contributions per 100 people (for only those DAs with at least one contribution)	$ 170.03	$ 157.68	$ 159.00

N: 56590 dissemination areas (DA).
Average total contributions (monetary and non-monetary) above $200 per 100 people in the DA. The prevalence of non-donors explains why means are lower than $200.

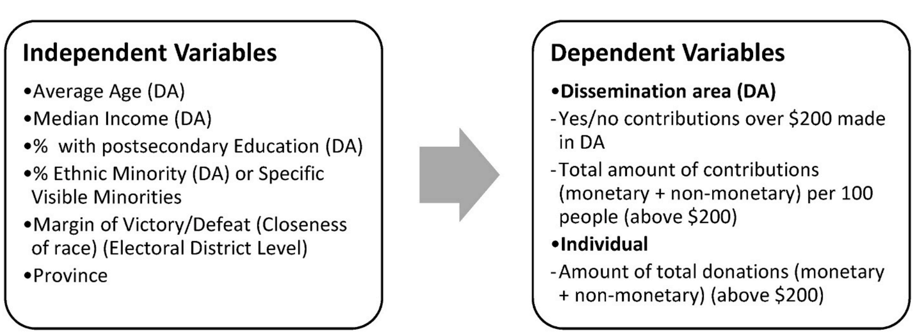

Fig. 2. Independent and dependent variables.

Table 2
Contributions and contributors (above $200).

Variable	2015		2019		2021	
	N	Mean	N	Mean	N	Mean
BY CONTRIBUTION						
Average Monetary Contribution	18,112	$ 585.20	10,564	$ 620.38	9137	$ 647.03
Average Non-Monetary Contribution	889	$ 435.99	999	$ 344.56	481	$ 587.11
Average Total Contributions	19,027	$ 577.43	11,710	$ 589.06	9775	$ 633.69
BY CONTRIBUTOR						
Average Monetary Contribution Total	16,950	$ 625.32	9967	$ 657.54	8794	$ 672.26
Average Non-Monetary Contribution Total	16,950	$ 22.87	9967	$ 34.55	8794	$ 32.11
Average Total Contribution	16,950	$ 648.19	9967	$ 692.07	8794	$ 704.37

Contribution limits to candidate (and EDA) were $1500 in 2015, $1600 in 2019 and $1650 in 2021. Note that candidates can donate more than these limits to their own campaign.

Table 3
Predicting whether at least one donor (above $200) resides in DA.

	(1)	(2)	(3)
	2015	2019	2021
Average Age	0.02***	0.02***	0.02***
	0.00	0.00	0.00
Median Income	0.00***	0.00***	0.00***
	0.00	0.00	0.00
Percent Postsecondary Educated	0.71***	0.22***	0.15**
	0.06	0.07	0.07
Percent Visible Minority	0.90***	0.65***	0.37***
	0.08	0.08	0.08
Margin of Victory	−0.07	0.06	0.27**
	0.08	0.10	0.11
NL	0.14	−0.28***	0.18*
	0.08	0.10	0.09
PE	1.54***	1.37***	1.62***
	0.13	0.13	0.13
NS	0.40***	0.61***	0.59***
	0.06	0.06	0.07
NB	−0.52***	−0.17**	−0.08
	0.08	0.08	0.09
QC	−1.05***	−0.92***	−1.35***
	0.03	0.04	0.05
MB	−0.51***	−0.46***	−0.48***
	0.06	0.07	0.08
SK	−0.34***	−0.40***	−0.03
	0.06	0.08	0.07
AB	−0.50***	−0.41***	−0.13***
	0.04	0.06	0.05
BC	−0.27***	−0.20***	−0.27***
	0.03	0.04	0.04
YT	0.70**	1.06***	2.05***
	0.31	0.31	0.31
NT	0.55**	−0.63*	0.26
	0.26	0.34	0.28
NU	−0.07	−0.28	−0.17
	0.50	0.61	0.61
_cons	−3.83***	−3.92***	−3.95***
	0.11	0.13	0.14
N	52414	52414	52414

Logistic Regression Analysis, Standard errors in parentheses ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.
Dissemination area is the unit of analysis. Note that 847 dissemination areas were attached to two or more federal electoral districts and are therefore dropped from analysis. Some census data at the dissemination area is not made public for privacy (ex. if the number of individuals is small and there is a risk of identifying information being released), thus there is a smaller number of DAs studied here.

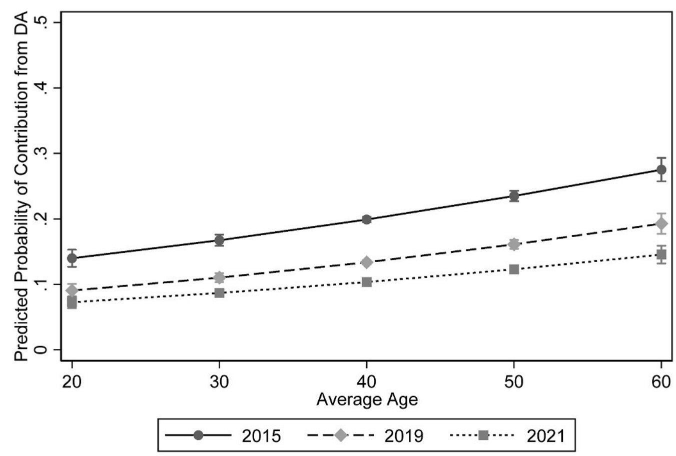

Fig. 3. Predicted probability of at least one donor (above $200) residing in DA, by average age.
Predicted probabilities (95% confidence intervals) from Table 3.
Unit of analysis is dissemination area (DA).

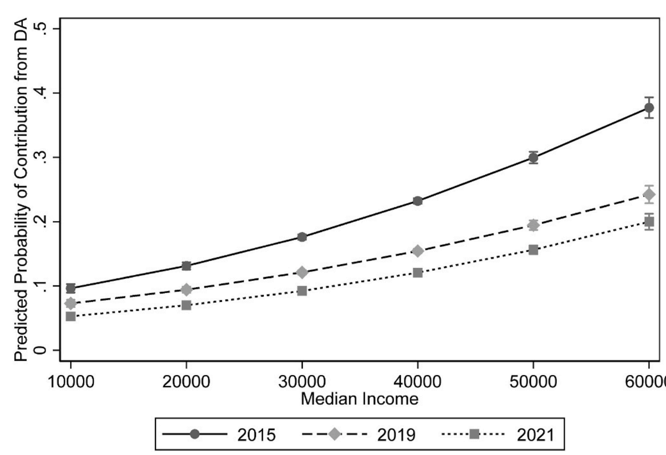

Fig. 4. Predicted probability of at least one donor (above $200) residing in DA, by median income.
Predicted probabilities (95% confidence intervals) from Table 3.
Unit of analysis is dissemination area (DA).

elections, the best source of information is available through the Elections Canada Political Financing Contributions database (Elections Canada, 2023b). Because of the transparency requirements required by law, Elections Canada regularly publishes an online database of campaign contributions, including donors' names, postal codes, contributions, and contribution type if the donor has cumulatively donated

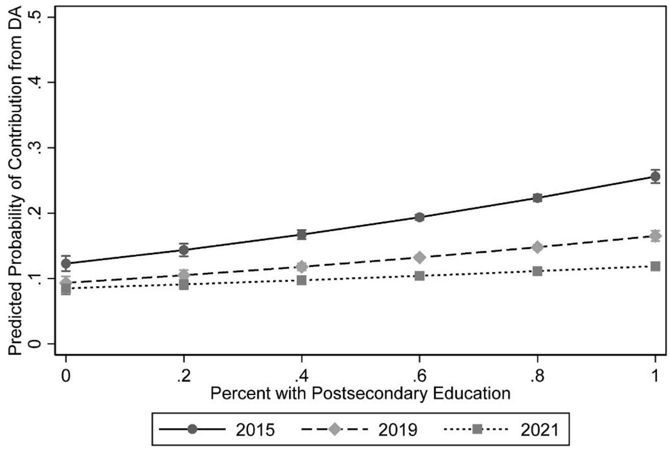

Fig. 5. Predicted probability of at least one donor (above $200) residing in DA, by post-secondary education.
Predicted probabilities (95% confidence intervals) from Table 3.
Unit of analysis is dissemination area (DA).

over $200 to a permitted entity (Elections Canada, 2023a).[1] Both monetary and non-monetary contributions are reported. It is important to note that this therefore does not include donations under $200.[2] This article considers only donations made to political candidates for the 2015, 2019 and 2021 general elections.[3]

To capture socio-demographic data, most previous work has used survey data. However, this comes with the challenges of representativeness in survey data, particularly, in this case, with getting access to the relatively small number of Canadians who are donors via traditional survey panels.[4] Other studies have extrapolated a donor's gender or ethnicity based on their name (Besco & Tolley, 2022; Tolley et al., 2022). However, this method is also problematic in that it makes assumptions regarding ethnic or gendered names.

This article instead leverages data from the Canadian census at the dissemination area level. The dissemination area is the smallest unit of reported census data; they aim to contain between 400 and 700 people in a small geographic area.[5] There are anywhere from 50 to about 250 dissemination areas in any electoral district. The 2016 census data is used in this article (Statistics Canada, 2019).[6] The Elections Canada database on contributions was matched with census data using the Postal Code Conversion File.[7] This file matches postal codes with census geographic areas. Further details on this process are found in the supplemental materials. The resulting dataset has individuals (donors) nested in dissemination areas, which are nested in electoral districts (see Fig. 1).[8]

Two main sets of analysis are conducted using these data (Fig. 2).

[1] "To each registered party, In total to all the registered associations, nomination contestants and candidates of each registered party, In total to all leadership contestants in a particular contest, To each independent candidate" (Elections Canada, 2023a).
[2] If there are not additional donations that would put the total over $200.
[3] Donations to registered parties and associations, and nomination contests are not studied here due to the local focus of this article.
[4] For work of this nature, see Garnett et al. (2022).
[5] For more, see Statistics Canada (2016).
[6] Additional census data was collected in 2021, but this article uses the 2016 census.
[7] Postal Code[OM] Conversion File (PCCF), 2017. Statistics Canada Catalogue no. 92-154-X. c Note: The process of linking postal codes with dissemination areas is not perfect. Postal codes were manually fixed when possible (e.g. swapping 0 zero for O when appropriate). Partial postal codes were matched where possible using weighted random selection of the postal codes with the same beginning characters (3 characters or more).

Table 4
Predicting whether at least one donor (above $200) resides in DA, with disaggregated Visible Minorities.

	(1)	(2)	(3)
	2015	2019	2021
Average Age	0.02***	0.02***	0.02***
	0.00	0.00	0.00
Median Income	0.00***	0.00***	0.00***
	0.00	0.00	0.00
Percent Postsecondary Educated	1.01***	0.70***	0.44***
	0.08	0.09	0.09
Percent South Asian	2.92***	1.76***	2.14***
	0.11	0.13	0.13
Percent Chinese	0.57***	0.08	−0.53***
	0.12	0.14	0.17
Percent Black	−2.46***	−1.32***	−1.93***
	0.28	0.31	0.36
Percent Filipino	−1.78***	−2.59***	−2.81***
	0.30	0.37	0.41
Percent Latin American	−3.02***	−2.11***	−1.81**
	0.61	0.69	0.77
Percent Arab	0.87**	1.06**	1.95***
	0.40	0.45	0.49
Percent Southeast Asian	0.77	0.29	−0.07
	0.58	0.68	0.76
Percent West Asian	−0.69	−0.01	−0.91
	0.59	0.67	0.79
Percent Korean	0.38	−0.44	0.56
	0.78	0.92	1.02
Percent Japanese	3.63***	0.10	−4.83***
	1.30	1.53	1.78
Margin of Victory	−0.04	0.07	0.20*
	0.09	0.10	0.12
NL	0.07	−0.31***	0.14
	0.08	0.10	0.10
PE	1.48***	1.34***	1.60***
	0.13	0.13	0.13
NS	0.38***	0.61***	0.58***
	0.06	0.06	0.07
NB	−0.56***	−0.19**	−0.09
	0.08	0.08	0.09
QC	−1.00***	−0.90***	−1.34***
	0.03	0.04	0.05
MB	−0.42***	−0.36***	−0.37***
	0.06	0.07	0.08
SK	−0.34***	−0.37***	0.01
	0.06	0.08	0.08
AB	−0.45***	−0.35***	−0.05
	0.04	0.06	0.05
BC	−0.37***	−0.22***	−0.26***
	0.04	0.04	0.05
YT	0.70**	1.10***	2.09***
	0.31	0.31	0.32
NT	0.60**	−0.54	0.35
	0.25	0.34	0.28
NU	−0.14	−0.30	−0.18
	0.50	0.61	0.61
_cons	−3.56***	−3.81***	−3.89***
	0.12	0.13	0.14
N	52414	52414	52414

Logistic Regression Analysis, Standard errors in parentheses ***p < 0.01, **p < 0.05, *p < 0.1.
Dissemination area is the unit of analysis. DAs that did not exactly match to electoral districts were dropped from analysis.

[8] Work of this nature, for example linking donations to geographic areas has taken place in an examination of municipal campaign finance in Seattle, where a small percentage of neighbourhoods held most campaign donors (Heerwig & McCabe, 2017).

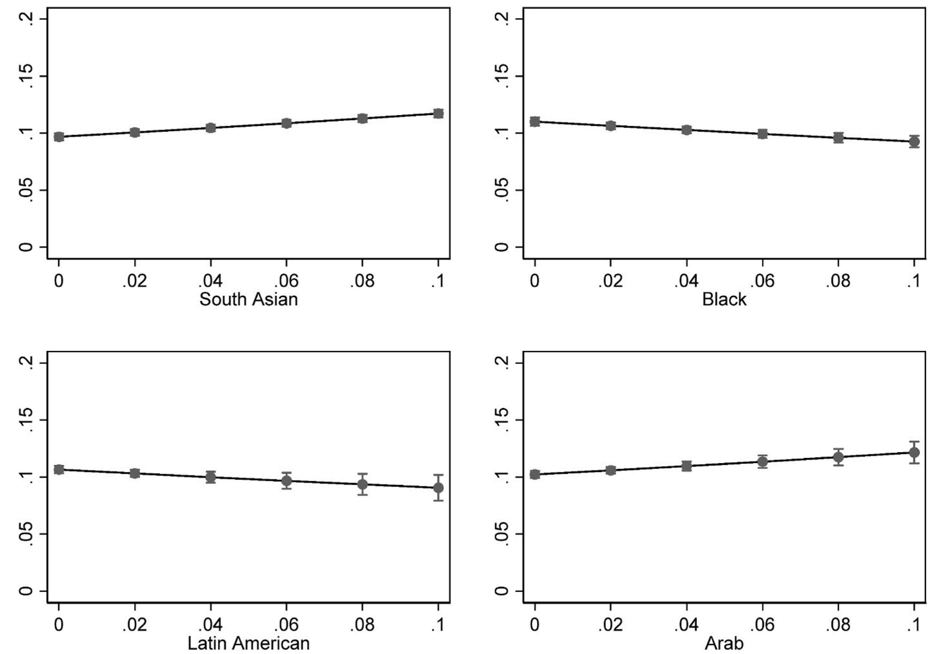

Fig. 6. Predicted probabilities of at least one donor residing in DA, by four visible minority groups (2021).
Predicted Probabilities (95% confidence intervals) from Table 4.
Unit of Analysis is Dissemination Area (DA). X axis is percent of population in DA who identify as that visible minority.

The first considers the ***dissemination area (DA)*** as the unit of analysis, collapsing the data to determine: firstly, whether there were any monetary contributions above $200 coming from the geographic area; and secondly, the total dollar amount of contributions coming from the dissemination area per 100 people (including both monetary and non-monetary contributions). These dependent variables are then matched with their respective census data (at the level of dissemination area). In these analyses, the margin of victory between the first and second placing candidates in the corresponding electoral district is used as a measure of competitiveness of the race.[9] Logistic or poisson regressions are used to model the relationship between DA-level characteristics and donations.

The second set of analysis considers the ***individual donor*** as the unit of analysis. Because the initial Elections Canada Political Financing Data are reported at the contributions level, these data are collapsed according to first name, last name, and postal code.[10] This gives a more accurate picture of the total dollar amount of donations per donor. Once again, these individual-level profiles are matched by postal code to census data in the donor's dissemination area.[11] For this analysis, the margin of defeat for the candidate to whom the contribution was made is used as a control variable to measure how close the race was, specifically for the individual donor's preferred candidate. Models are therefore logistic or poisson regressions with fixed effects, with DAs nested in electoral districts. For these models, rather than margin of victory, margin of defeat is used as a control variable. This variable captures how competitive the race is for each donor, based on how close the contest was for the candidate they donated to. In other words, margin of defeat is 0 if the donor's candidate won and increases as the donor's candidate's vote margin gets further from the eventual winner in that electoral district. The higher the number, the less competitive their candidate. Multi-level OLS regression models with donors nested in electoral districts are used to model the predictors of donation levels for donors. Summary data by dissemination area and fixed effects by district is found in Appendix 1.

4. Results: Summary of donations to candidates in 2015, 2019 and 2021

What do the data show us about donations to candidates in general? The first major story here is that donations to local candidates are on the decline (see Table 1). Considering whether *any* contributions to district-level candidates were made in a dissemination area, we see that of the 56590 DAs, 22% had contributions from 2015, which nearly halved to about 12% in 2021. Similarly, we see the average amount of contributions in a DA per 100 people decline over the same period, from $170 in 2015 to about $160 in 2021.

The same decline in donations to district-level candidates is found in individual-level data on contributors (Table 2). The number of contributions above $200 made in 2015 were over 18,000, which is about double the number of contributions made in 2021 (9137). However, at the same time, the average amount of each contribution went up, from an average contribution amount of nearly $600 in 2015 to nearly $650 in 2021. When the data are collapsed by first name, last name, and postal code to estimate the number of unique donors, a similar picture develops.[12] The number of unique donors has declined between 2015 and 2021, but the average amount any unique donor contributes to district-level candidates has increased.

Why is there a noticeable decline in number of contributions and contributors to district-level candidates, but at the same time an increase in contribution amounts? This could be due to several factors. Regarding the number of contributions, interest in the campaigns may have

[9] Note that when matching DAs to Electoral Districts, there was not a perfect match for some DAs. These were dropped in the regression analysis.

[10] For example, an individual could make two $100 donations, but each are recorded individually. Note that there remains a small chance that due to typographical errors or if there were two people living in the same household with the same names, the collapse of data could miss some repeat donors.

[11] Research using a gender or ethnicity API has been conducted by Besco and Tolley (2022), Tolley, Besco, and Sevi (2022) and McMahon et al. (2021).

[12] Note that data entry errors or typos may result in an overestimate of unique donors. Likewise, if people with the same name exist at the same postal code, this may underestimate unique donors.

Table 5
Total amounts of candidate contributions (above $200) made in the DA per 100 people (DA Level).

	(1)	(2)	(3)
	2015	2019	2021
Average Age	0.03***	0.03***	0.03***
	0.00	0.00	0.00
Median Income	0.00***	0.00***	0.00***
	0.00	0.00	0.00
Percent Postsecondary Educated	0.68***	0.12***	0.03***
	0.00	0.00	0.01
Percent Visible Minority	0.15***	0.20***	−0.03***
	0.00	0.00	0.01
Margin of Victory	−0.01	−0.10***	0.18***
	0.01	0.01	0.01
NL	0.44***	−0.05***	0.44***
	0.00	0.01	0.01
PE	1.27***	1.02***	1.47***
	0.01	0.01	0.01
NS	0.46***	0.49***	0.49***
	0.00	0.00	0.00
NB	−0.34***	−0.01**	−0.11***
	0.01	0.01	0.01
QC	−0.75***	−0.74***	−1.07***
	0.00	0.00	0.00
MB	−0.32***	−0.44***	−0.53***
	0.00	0.01	0.01
SK	−0.12***	−0.23***	0.17***
	0.00	0.01	0.01
AB	−0.36***	−0.22***	−0.04***
	0.00	0.00	0.00
BC	−0.18***	−0.16***	−0.14***
	0.00	0.00	0.00
YT	0.29***	0.64***	2.03***
	0.02	0.02	0.01
NT	0.51***	−0.64***	−0.21***
	0.01	0.03	0.02
NU	−0.37***	−1.13***	−0.97***
	0.04	0.07	0.08
_cons	0.80***	0.89***	0.52***
	0.01	0.01	0.01
N	52414	52414	52414

Poisson Regression Analysis, Standard errors in parentheses ***p < 0.01, **p < 0.05, *p < 0.1.
Dissemination area is the unit of analysis. DAs that did not exactly match to electoral districts were dropped from analysis.

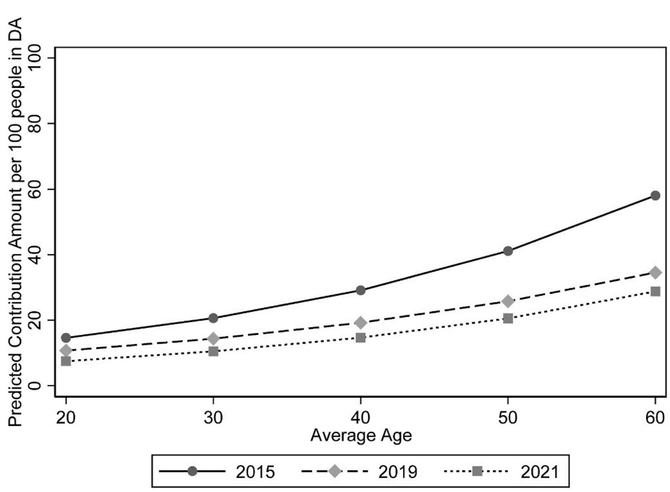

Fig. 7. Predicted amounts of candidate donations (above $200) per 100 people in DA, by average age.
Predictive margins (95% confidence intervals) from Table 5. Unit of analysis is dissemination area (DA).

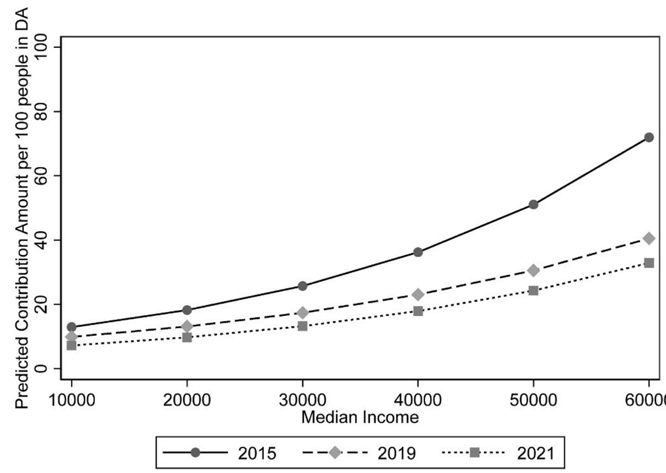

Fig. 8. Predicted probability of total amounts of candidate donations (above $200) per 100 people in DA, by median income.
Predictive margins (95% confidence intervals) from Table 5. Unit of analysis is dissemination area (DA).

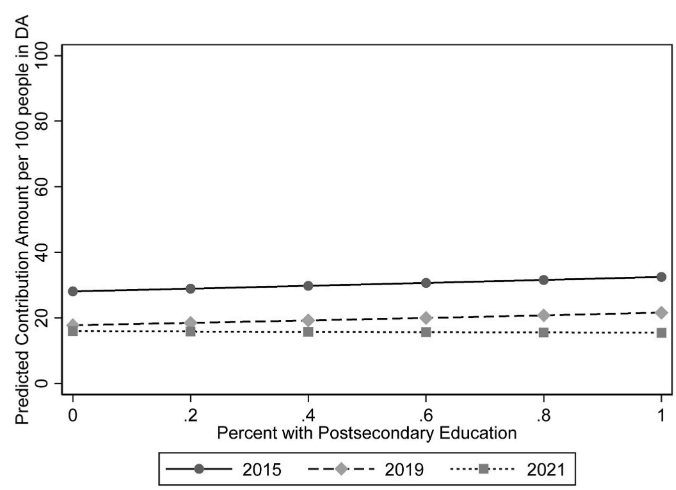

Fig. 9. Predicted probability of total amounts of candidate donations (above $200) per 100 people in DA, by post-secondary education.
Predicted margins (95% confidence intervals) from Table 5.
Unit of analysis is dissemination area (DA).

dropped between 2015 and 2021. The 2015 election was particularly contentious with three parties in the running to form the government, potentially attracting more donors to local contests. There was considerably less excitement (as evidenced in part by some decline in voter turnout) for the 2019 and 2021 elections (Elections Canada, 2023c). A decline in the number of contributions in 2021 is likewise unsurprising, given the financial impact of the COVID-19 pandemic.

Another reason for a decline in contributions to district-level candidates could be a diverting of funds from local candidates to electoral district associations or the central party. In the wake of the 2015 election, where major parties attracted new supporters, new donors may have moved their donations to electoral district associations or central parties. Additionally, in the decline of the per-vote subsidy, parties needed to replace the revenue with individual donors who may have diverted their funds from candidates.

However, the increase in average contribution amounts also makes sense. Contribution limits increased over the same period of time, from $1500 in 2015 to $1650 in 2021, making the limit of donations $150 higher from the first election studied here to the last. Campaigns must

Table 6
Total amount of candidate contributions (above $200) made in the DA per 100 people (DA Level), disaggregated ethnicity.

	(1)	(2)	(3)
	2015	2019	2021
Average Age	0.03***	0.03***	0.03***
	0.00	0.00	0.00
Median Income	0.00***	0.00***	0.00***
	0.00	0.00	0.00
Percent Postsecondary Educated	0.18***	0.22***	0.02***
	0.00	0.00	0.01
Percent South Asian	2.50***	1.73***	2.05***
	0.01	0.01	0.01
Percent Chinese	0.86***	0.11***	−0.52***
	0.01	0.01	0.01
Percent Black	−2.95***	−1.72***	−2.17***
	0.02	0.03	0.03
Percent Filipino	−1.39***	−3.16***	−2.64***
	0.02	0.03	0.03
Percent Latin American	−3.59***	−2.02***	−4.58***
	0.05	0.05	0.07
Percent Arab	−0.70***	0.86***	0.59***
	0.03	0.03	0.04
Percent Southeast Asian	0.94***	1.13***	0.43***
	0.04	0.05	0.06
Percent West Asian	−0.94***	−0.80***	−0.46***
	0.04	0.05	0.06
Percent Korean	−1.11***	−2.86***	−0.76***
	0.06	0.08	0.08
Percent Japanese	6.59***	0.77***	−1.22***
	0.08	0.11	0.13
Margin of Victory	0.04***	−0.10***	0.10***
	0.01	0.01	0.01
NL	0.36***	−0.08***	0.40***
	0.00	0.01	0.01
PE	1.20***	1.00***	1.44***
	0.01	0.01	0.01
NS	0.43***	0.49***	0.48***
	0.00	0.00	0.00
NB	−0.39***	−0.03***	−0.13***
	0.01	0.01	0.01
QC	−0.66***	−0.72***	−1.02***
	0.00	0.00	0.00
MB	−0.26***	−0.34***	−0.44***
	0.00	0.01	0.01
SK	−0.14***	−0.20***	0.20***
	0.00	0.01	0.01
AB	−0.33***	−0.16***	0.03***
	0.00	0.00	0.00
BC	−0.34***	−0.19***	−0.18***
	0.00	0.00	0.00
YT	0.25***	0.67***	2.04***
	0.02	0.02	0.01
NT	0.55***	−0.55***	−0.15***
	0.01	0.03	0.02
NU	−0.45***	−1.15***	−1.00***
	0.04	0.07	0.08
_cons	1.10***	1.00***	0.64***
	0.01	0.01	0.01
N	52414	52414	52414

Poisson Regression Analysis, Standard errors in parentheses ***p < 0.01, **p < 0.05, *p < 0.1.
Dissemination areas is the unit of analysis. DAs that did not exactly match to electoral districts were dropped from analysis.

keep up with inflation and new costs to elections and continue to require donors to pay their bills. Thus, when unable to attract large numbers of new donors, due to interest in the campaign or social pressures like the pandemic, candidates relied on pulling more from their smaller number of donors.

5. Results: Geographic predictors of donations

The models in Table 3 use the DA as the unit of analysis, with a binary variable of whether any contributions to district-level candidates came from that DA.

Starting with ***age***, we note that DAs with a higher average age are more likely to have at least one donor living in them (Table 3). This reflects previous work on age and political participation, that shows that traditional forms of participation are more common among older citizens. Fig. 3 presents the predicted probability of a DA having at least one donor, based on the average age in that DA. It can be noted that the magnitude of this effect is only about a 10-percentage point difference between an average age of 20 and an average age of 60 in 2015, and far less in 2021. However, since age is measured at the level of the DA, this presents perhaps a more conservative picture of the impact of age.

As expected, median ***income*** is also a statistically significant predictor of whether there are any donors in a DA, pointing to a conclusion that donors are more commonly drawn from richer areas. In Fig. 4 we see one of the steepest increases is predicted probability of having at least one donor in a DA. In 2015, the average median income among DAs was about $35,000. The predicted probability of having at least one donor in a DA with this average median income was about 17%. However, if the median income increases $10,000, the predicted probability jumps about 5 percentage points to 22%. If it drops $10,000, it decreases about 5 percentage points to 12%. The income of an area therefore makes a big difference in whether there are any donors living there.

Turning to ***education*** levels, we note that having postsecondary education has a positive impact on the likelihood of there being donors residing in an area. The magnitude of this relationship is most pronounced in 2015, where the difference between a DA with 10% postsecondary education and 50% postsecondary education is about four percentage points, and much less pronounced in 2021, where the same jump results in less than half a percentage point change in predicted probability of having donors in the DA (Fig. 5).

Table 3 also shows that the likelihood of contribution increases as the percentage *of* ***visible minorities*** in the DA increases. To better understand this finding, Table 4 presents the same models with disaggregated visible minorities. In line with previous findings from the Canadian context, we note that disaggregating the visible minorities presents a more nuanced picture of donation levels in these minority communities. DAs with a greater percentage of South Asian, South-East Asian, and Arab visible minorities are more likely to give contributions to candidates, while DAs with a greater percentage of Black, Filipino, and Latin American minorities are less likely to give contributions to candidates.

To illustrate the magnitude of these effects, consider the examples of four visible minority groups in 2021 in Fig. 6. The predicted probability of having at least one donor in the DA increases as the percentage of South Asians or Arabs in the DA increases. Conversely, however, the chances of having at least one donor decreases as the percentage of Black people or Latin Americans in the DA increases. While these effects are small compared to those of median income, for example, they remain greater than the same proportionate impact of postsecondary education in 2021.

These findings lend support to the findings of Besco and Tolley (2022) in the Canadian context, that shows that some groups of Asian Canadians are in fact more likely to donate. It also shows that the findings of previous work in the American context, showing that Black and Latin Americans are less likely to donate, is supported in the Canadian context as well (King, 2009). Furthermore, disaggregation here gives a more nuanced picture than the initial models with visible minorities grouped together do. Importantly, these findings demonstrate the importance of disaggregating visible minority communities: they have different patterns of political participation, especially when it comes to donating.

6. Results: How much is donated?

Can these data also reveal something about the differences in how much is being donated? Table 5 presents as a dependent variable the

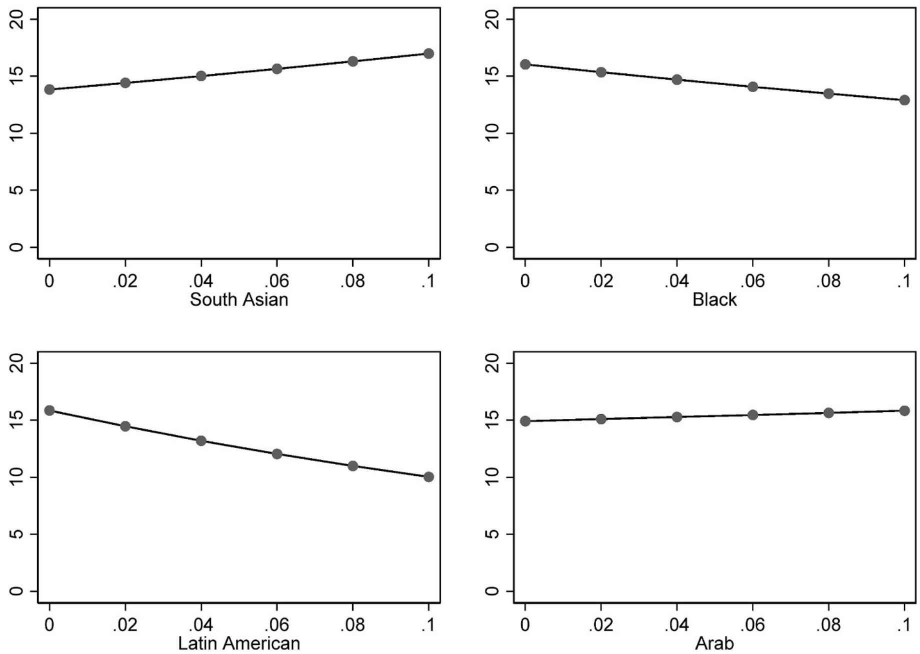

Fig. 10. Predicted probabilities of total amounts of candidate donations (above $200) per 100 people in DA, by four visible minority groups (2021).
Predictive margins (95% confidence intervals) from Table 6.
Unit of analysis is dissemination area (DA). X axis is percent of population in DA who identify as that visible minority. Y axis is in CAD $.

total amounts of candidate contributions made in each DA, per 100 people (to account for varying DA populations). In this case, a poisson regression is used since there are many DAs in which there are no donors.

The results follow a similar pattern to the predictors of the presence of donors in a DA. Higher average age and higher median income are positively related to higher donation levels in the DA. For example, in 2015, the predicted total amounts of contributions per 100 people in a DA jumps about $20 from a DA with an average age of 20 to a DA with an average age of 60 (see Fig. 7).

The predicted total contributions (per 100 people) likewise increases as the median income increases. The increase in predicted contribution amount per 100 people again jumps from $10 for a DA with a median income of $20,000 to $24 for a DA of $50,000 (in 2021) (see Fig. 8). Thus, we can see that income, unsurprisingly, not only influences the likelihood of donating, but also the amounts of those donations.

We see less of a stark trend for the amount donated by education. Post-secondary education appears to have a much more pronounced effect on whether anyone in the DA donated, than on the amounts of those contributions. In fact, the coefficient in 2021 is negative. To make sense of this finding, we might suggest then that education influences who donates, but not necessarily how much they donate. Perhaps those with post-secondary education are in the types of networks that make them likely to be asked, or have the political sophistication necessary to participate via this means, but the pure effects of education on donating do not extend to the amounts they donate like the variables of age and income do (see Fig. 9).

The percent of visible minorities in a dissemination area is not consistently positively or negatively related to donation levels (the coefficients are positive in 2015 and 2019, and negative in 2021, see Table 5). However, disaggregating ethnic minority groups in Table 6, we see similar trends as the previous analysis of these groups. DAs with a higher percentage of South Asians and South-East Asians (and in all but 2021, Chinese) have higher total contribution amounts (per 100 people) in the DA (see also Fig. 10). At the same time, those DAs with greater percentages of Black people, Filipinos, Latin Americans, and West Asians see lower donation totals. Again, this supports suggestions that some visible minority communities are more prime to donate, be that because of candidate networks or other avenues of influence. But as in other research (for example, Besco and Tolley (2022)), the amounts of these donations is not substantially larger (only a few dollars more) in areas with larger specific visible minority groups who tend to donate more (like South Asian or Southeast Asian).

A final analysis on whether socio-demographic characteristics can predict donation amounts considers only donors and compares them to the amounts of their contributions (Table 7 and Fig. 11) and the socio-demographic profile of the DA in which they are living.

The results, however, show us that when just taking donors into consideration, the socio-demographic profile of the area where they reside is a poor predictor of their donation amount. To understand this another way, it appears that when non-donors (the zeros) are removed from the dataset, socio-demographic profiles cease to be a good way of predicting how much an individual donor will give. This shows that the biggest socio-demographic differences found in these data of donors of more than $200 are whether or not contributions are made, rather than the amount of these contributions.

7. Conclusion

This article combined data from the Elections Canada Political Financing Contributions database with data from 2016 census at the Dissemination Area level to gauge the socio-demographic predictors of who donates to district-level candidates and how much. This research considers data primarily at the level of dissemination area, capturing the socio-demographic profiles of the geographic areas where donors live. It is unique in that it does not rely on a sample of donors, survey data, nor on extrapolation of donor characteristics by the donor's name. Instead, the data on donors presented are the full list of contributors ($200+), and is not threatened by small sample size, survey response bias, or the potential inaccuracies of determining socio-demographic characteristics based on name.

Like previous studies from the American context, this article finds that the areas that are most likely to have donors have a higher average age, a higher median income, and a higher percentage of the population with post-secondary education. Income, unsurprisingly, has the most substantial effect of these variables. These findings support previous

Table 7
Total contribution amounts (above $200) (individual level).

	(1)	(2)	(3)
	2015	2019	2021
Average Age	−0.56	−0.13	−0.14
	0.86	1.59	1.19
Median Income	0.00***	0.00**	0.00**
	0.00	0.00	0.00
Percent Visible Minority	68.75**	63.17	45.79
	34.24	56.21	45.13
Percent Postsecondary Educated	12.08	−103.89	45.63
	48.64	91.74	77.87
Margin of Defeat	−106.24**	−55.02	−454.74***
	49.44	94.49	85.10
NL	154.37**	45.34	−103.31
	67.20	54.70	106.39
PE	−49.12	−1.84	−239.72
	66.48	63.77	362.20
NS	93.90	27.52	−236.24*
	62.40	74.20	125.46
NB	101.04	189.73***	−76.87
	120.77	58.34	89.68
QC	30.65	104.36	20.27
	39.83	66.14	81.14
MB	−29.77	−68.37	−113.03
	72.94	89.66	93.65
SK	−70.63	−26.80	−66.19
	89.26	91.77	68.88
AB	42.85	−18.05	−55.58
	39.80	61.63	55.35
BC	60.72	4.66	−62.57
	65.04	48.47	62.02
YT	225.25***	−132.31	153.49
	20.31	119.31	110.28
NT	44.47	−43.78	−145.72***
	298.83	50.62	39.64
NU	549.01***	−1090.58***	−666.86***
	92.15	39.07	198.95
_cons	491.94***	540.12***	674.15***
	47.82	102.05	76.89
N (individuals)	18207	9813	9296
N (electoral districts)	337	328	333
R-sq	0.00	0.00	0.02

Multi-level OLS Regression Analysis (Individual Contributors nested in electoral districts).
Robust standard errors in parentheses ***p < 0.01, **p < 0.05, *p < 0.1.
Note: Some loss of data where census data was not reported for privacy concerns.

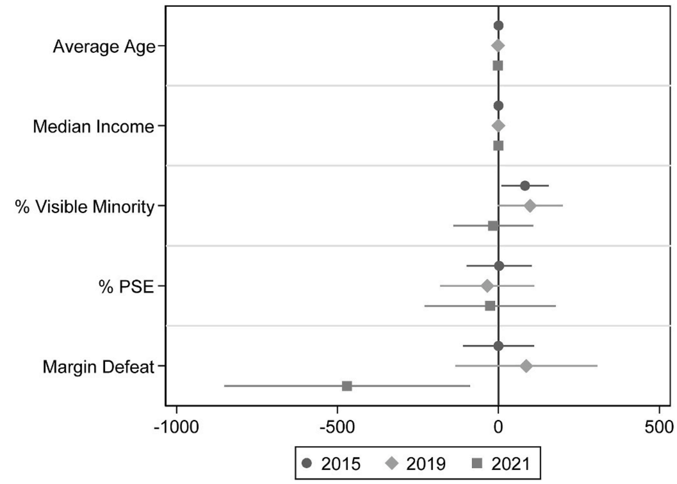

Fig. 11. Coefficient plot from Table 7.
PSE = Post-secondary Education.

international work that suggests the characteristics of those most likely to engage in traditional forms of participation (such as voting) are also most likely to donate to campaigns. It shows that even in a country like Canada, where there are strict regulations, transparency, and caps on campaign donations, these policies do not altogether eliminate the inequalities of voice posed by campaign donations.

The data further supports previous findings from the Canadian context, using other methodologies, suggesting that visible minorities of South and Southeast Asian origin are most likely to donate to local campaigns and reflects American research that Black and Latin Americans are less likely to donate. It also highlights the need for future research to ensure that visible minority groups are not simply 'lumped together' in the study of their participation habits.

One unique finding of this paper, that is impossible to capture with survey data asking simply whether an individual has donated or not (the question commonly asked in Canadian election studies), is that the effects of these socio-demographic variables are somewhat weakened (though not eliminated) when the total contributions amounts are considered. This points to a major difference between donors and non-donors, rather than between the amounts of donations made by individuals. In other words, when considering contributions at the $200+ level (those that are reported publicly by Elections Canada) the key difference appears to be whether an individual gave, not the amount of this donations.

There remain some limitations to this study, which are also potential areas for further research. Firstly, information aggregated to the DA does not have much ability to speak to issues of gender differences in donating, nor to the attitudinal or ideological predictors of donating. Additionally, it cannot speak to the motivations of why individuals would donate to a campaign compared with donating to a central or district-level political party. These questions are outside of the scope of this article and it would be better to rely on other sources of data.

Funding

This research was funded by SSHRC Insight Grant - 435 2019 0065.

CRediT authorship contribution statement

Holly Ann Garnett: Writing – review & editing, Writing – original draft, Visualization, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

The author acknowledges the research assistance of Sean Grogan and Madison MacGregor.

Appendix 1. Summary Statistics (by Dissemination Area)

2015	Mean	Std Dev	Max
Number of Contributions	0.37	1.05	59.00
Number of Contributions per 100 people	0.01	0.05	2.50
Average Monetary Contribution Amount	$ 122.05	$ 323.73	$ 5000.00
Average Non-Monetary Contribution Amount	$ 6.31	$ 88.88	$ 5000.00
Average Monetary Contribution Amount per 100 people	$ 34.68	$ 213.17	$ 41,000.00
Average Non-Monetary Contribution Amount per 100 people	$ 1.98	$ 34.86	$ 4207.32
2019	Mean	Std Dev	Max
Number of Contributions	0.23	0.94	101.00
Number of Contributions per 100 people	0.01	0.04	1.16
Average Monetary Contribution Amount	$ 86.78	$ 298.93	$ 7047.44
Average Non-Monetary Contribution Amount	$ 5.15	$ 102.63	$ 11,448.36
Average Monetary Contribution Amount per 100 people	$ 21.55	$ 107.82	$ 10,000.00
Average Non-Monetary Contribution Amount per 100 people	$ 1.44	$ 29.91	$ 2936.73
2021	Mean	Std Dev	Max
Number of Contributions	0.18	0.64	19.00
Number of Contributions per 100 people	0.00	0.03	5.00
Average Monetary Contribution Amount	$ 75.46	$ 286.88	$ 5000.00
Average Non-Monetary Contribution Amount	$ 3.07	$ 62.60	$ 4494.15
Average Monetary Contribution Amount per 100 people	$ 18.38	$ 93.01	$ 6600.00
Average Non-Monetary Contribution Amount per 100 people	$ 0.83	$ 18.01	$ 1754.72

Note: Because the number of people in a DA may be very small, the maximum may be above the limit.

Appendix 2. Tables 3 and 4 with Multi-level models

Table 3
with Multi-Level Models
Dependent Variable: Whether there were any contributions from the DA

	(1)	(2)	(3)
	2015	2019	2021
Average Age	0.03***	0.03***	0.04***
	0.00	0.00	0.00
Median Income	0.00***	0.00***	0.00***
	0.00	0.00	0.00
Percent Visible Minority	0.73***	0.61***	0.71***
	0.01	0.01	0.01
Percent Postsecondary Educated	0.12***	0.16***	0.04***
	0.00	0.00	0.00
N (DAs)	52414	52414	52414
N (EDs)	338	338	338

Multi-level Logistic Regression Analysis (Dissemination areas nested in electoral districts), with district fixed effects.
DAs that did not exactly match to electoral districts were dropped from analysis.
Standard errors in parentheses ***p $<$ 0.01, **p $<$ 0.05, *p $<$ 0.1.

Table 4
with Multi-Level Models
Dependent Variable: Total amounts of donations in DA per 100 people

	(1)	(2)	(3)
	2015	2019	2021
Average Age	0.02***	0.03***	0.02***
	0.00	0.00	0.00
Median Income	0.00***	0.00***	0.00***
	0.00	0.00	0.00
Percent Visible Minority	1.38***	1.19***	1.16***
	0.10	0.12	0.13
Percent Postsecondary Educated	0.81***	0.60***	0.50***
	0.09	0.09	0.10
N (DAs)	52414	52414	52414
N (EDs)	338	338	338

Multi-level OLS Regression Analysis (Dissemination areas nested in electoral districts), with district fixed effects.

DAs that did not exactly match to electoral districts were dropped from analysis.
Standard errors in parentheses ***p < 0.01, **p < 0.05, *p < 0.1.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.polgeo.2024.103077.

References

Ansolabehere, S., John, M., de Figueiredo, & Snyder, J. M. (2003). Why is there so little money in U.S. Politics? *The Journal of Economic Perspectives, 17*(1), 105–130. http://www.jstor.org/stable/3216842.

Bartels, L. M. (2008). *Unequal democracy : The political economy of the new gilded age*. Princeton: Princeton University Press.

Bartels, L. M. (2009). Economic inequality and political representation. In J. Lawrence, & K. Desmond (Eds.), *The unsustainable American state*. New York: Oxford University Press.

Beetham, D. (Ed.). (1994). *Defining and measuring democracy*. London: Sage Publications.

Besco, R., & Tolley, E. (2022). Ethnic group differences in donations to electoral candidates. *Journal of Ethnic and Migration Studies, 48*(5), 1072–1094. https://doi.org/10.1080/1369183X.2020.1804339

Boese, V. A., & Wilson, M. C. (2022). Contestation and participation: Concepts, measurement, and inference. *International Area Studies Review, 26*(2), 89–106. https://doi.org/10.1177/22338659221120970, 10.1177/22338659221120970.

Carmichael, B., & Howe, P. (2014). Political donations and democratic equality in Canada. *Canadian Parliamentary Review, 37*(1), 1.

Cho, W. T. (2002). Tapping motives and dynamics behind campaign contributions: Insights from the Asian American case. *American Politics Research, 30*(4), 347–383. https://doi.org/10.1177/1532673X02030004001, 10.1177/1532673X02030004001.

Coppedge, M., Alvarez, A., & Maldonado, C. (2008). Two persistent dimensions of democracy: Contestation and inclusiveness. *The Journal of Politics, 70*(3), 632–647. https://doi.org/10.1017/S0022381608080663

Cordova, A., & Rangel, G. (2017). Addressing the gender gap: The effect of compulsory voting on women's electoral engagement. *Comparative Political Studies, 50*(2), 264.

Costantini, E., & King, J. (1982). Checkbook democrats and their copartisans: Campaign contributors and California political leaders, 1964-1976. *American Politics Research, 10*(1), 65–92.

Cross, W., & Young, L. (2006). Are Canadian political parties empty vessels? Membership, engagement and policy capacity. *Institute for Research on Public Policy Choices, 12*(4).

Dahl, R. A. (1971). *Polyarchy: Participation and opposition*. New Haven and London: Yale University Press.

Eagles, M. (2004). The effectiveness of local campaign spending in the 1993 and 1997 federal elections in Canada. *Canadian Journal of Political Science, 37*(1), 117–136. http://www.jstor.org/stable/25165603.

Elections Canada. (2021). *Political financing handbook for candidates and official agents*. https://www.elections.ca/pol/can/man/ec20155_c76/2021-04_e.pdf?ver=1.1.

Elections Canada. (2023a). *Limits on contributions – 2023*. https://www.elections.ca/content.aspx?section=pol&dir=lim&document=lim2023&lang=e.

Elections Canada. (2023b). *Political financing*. https://www.elections.ca/content.aspx?section=fin&&document=index&lang=e.

Elections Canada. (2023c). *Voter turnout at federal elections and referendums*. https://www.elections.ca/content.aspx?section=ele&dir=turn&document=index&lang=e.

Everson, D. H. (1982). The decline of political parties. *Proceedings of the Academy of Political Science, 34*(4), 49–60.

Ferreira, V. (2015). "Canadian election 2015: Poll tracker." the national post Accessed 12-19 https://nationalpost.com/news/politics/canadian-election-2015-poll-tracker.

Francia, P. L., Green, J. C., Herrnson, P. S., Powell, L. W., & Wilcox, C. (2003). *The financiers of congressional elections: Investors, ideologues, and intimates*. New York: Columbia University Press.

Garnett, H. A., Scott, P., Young, L., & Cross, W. P. (2022). Lifeblood of the party: Motivations for political donations in Canada. *American Review of Canadian Studies, 52*(4), 422–445. https://doi.org/10.1080/02722011.2022.2147756

Gilens, M., & Page, B. I. (2014). Testing theories of American politics: Elites, interest groups, and average citizens. *Perspectives on Politics, 12*(3), 564–581, 10.1017/S1537592714001595 https://www.cambridge.org/core/product/62327F513959D0A304D4893B382B992B.

Gimpel, J. G., Lee, F. E., & Kaminski, J. (2006). The political geography of campaign contributions in American politics. *The Journal of Politics, 68*(3), 626–639.

Grumbach, J. M., Alexander, S., & Staszak, S. (2022). Gender, race, and intersectionality in campaign finance. *Political Behavior, 44*(1), 319–340. https://doi.org/10.1007/s11109-020-09619-0, 10.1007/s11109-020-09619-0.

Harell, A. (2009). Equal participation but separate paths?: Women's social capital and turnout. *Journal of Women, Politics & Policy, 30*(1), 1–22.

Heerwig, J., & McCabe, B. J. (2017). High-dollar donors and donor-rich neighborhoods: Representational distortion in financing a municipal election in Seattle. *Urban Affairs Review, 55*(4), 1070–1099. https://doi.org/10.1177/1078087417728378, 10.1177/1078087417728378.

Jacobson, G. C. (1980). *Money in congressional elections*. New Haven: Yale University Press.

James, R. N. (2009). An econometric analysis of household political giving in the USA. *Applied Economics Letters, 16*(5), 539–543.

Johnson, B. (2013). *Political giving: Making sense of individual campaign contributions, Book*. Whole. Boulder, Colorado: First Forum Press.

Jones, R. S. (1990). Contributing as participation. In N. Margaret Latus, & J. R. Johannes (Eds.), *Money, elections, and democracy: Reforming congressional campaign finance*. Boulder: Westview Press.

King, M. (2009). Reluctant donors: African Americans, campaign contributors, and the obama effect-or lack of it. *Souls, 11*(4), 389–406.

Lipsitz, K., & Panagopoulos, C. (2011). Filled coffers: Campaign contributions and contributors in the 2008 elections. *Journal of Political Marketing, 10*(1–2), 43–57. https://doi.org/10.1080/15377857.2011.540193

Mateos, P. (2007). A review of name-based ethnicity classification methods and their potential in population studies. *Population, Space and Place, 13*(4), 243–263, 10.1002/pspd.457 https://onlinelibrary.wiley.com/doi/abs/10.1002/psp.457.

McMahon, N., Sayers, A., & Alcantara, C. (2021). Political donations and the gender gap during COVID-19. *Party Politics, 29*(1), 176–184. https://doi.org/10.1177/13540688211047768

Meisel, J., & Mendelsohn, M. (1996). Meteor? Phoenix? Chameleon? The decline and transformation of party in Canada. In H. G. Thorburn (Ed.), *Party politics in Canada*. Scarorough: Prentice Hall Canada Inc.

Norris, P. (2002). *Democratic Phoenix: reinventing political activism*. New York: Cambridge University Press.

Schlozman, K. L., Verba, S., Henry, E., & Brady. (2012). *The unheavenly chorus: Unequal political voice and the broken promise of American democracy*. Princeton: Princeton University Press.

Smets, K., & van Ham, C. (2013). The embarrassment of riches? A meta-analysis of individual-level research on voter turnout. *Electoral Studies, 32*(2), 344–359.

Statistics Canada. (2016). Dissemination area (DA). In *Dictionary, census of population, 2016*.

Statistics Canada. (2019). *Individuals file, 2016 census of population*. Public Use Microdata Files), Article 2016001.

Tolley, E., Besco, R., & Sevi, S. (2022). Who controls the pruse strings? A longitudinal study of gender and donations in Canadian politics. *Politics and Gender, 18*(1), 244–272. https://doi.org/10.1017/S1743923X20000276

Verba, S., Burns, N., & Lehman Schlozman, K. (1997). Knowing and caring about politics: Gender and political engagement. *The Journal of Politics, 59*(4), 1051–1072. http://www.jstor.org/stable/2998592.

Verba, S., Lehman Schlozman, K., & Henry, B. (1995). *Voice and equality: Civic voluntarism in American politics*. Cambridge: Harvard University Press.

Wattenberg, M. P. (1984). *The decline of American political parties 1952-1980*. Cambridge: Harvard University Press.

Winters, J. A., & Page, B. I. (2009). Oligarchy in the United States? *Perspectives on Politics, 7*(4), 731–751. https://doi.org/10.1017/S1537592709991770. proxy.queensu.ca/login?url https://www.proquest.com/scholarly-journals/oligarchy-united-states/docview/869995258/se-2?accountid=6180.

Wong, M. Y. H. (2021). Democracy, hybrid regimes, and inequality: The divergent effects of contestation and inclusiveness. *World Development, 146*, Article 105606. https://doi.org/10.1016/j.worlddev.2021.105606